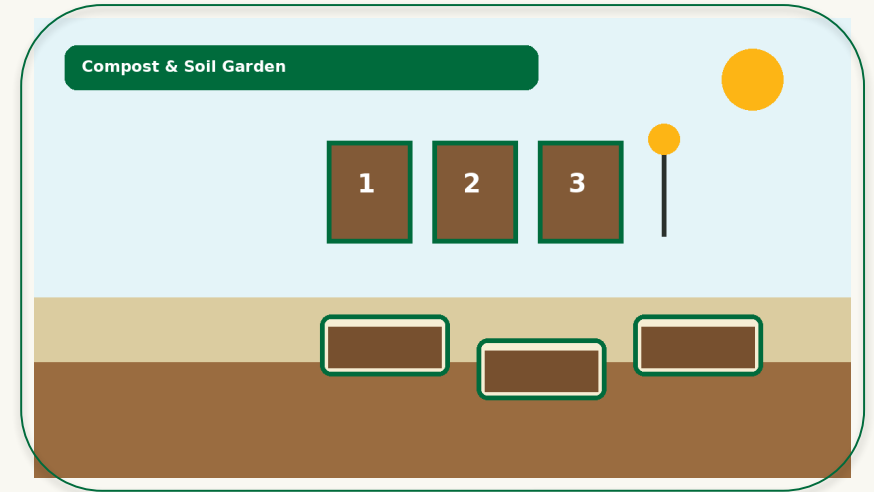


Compost & Soil Garden

Soil Health and Organic Matter

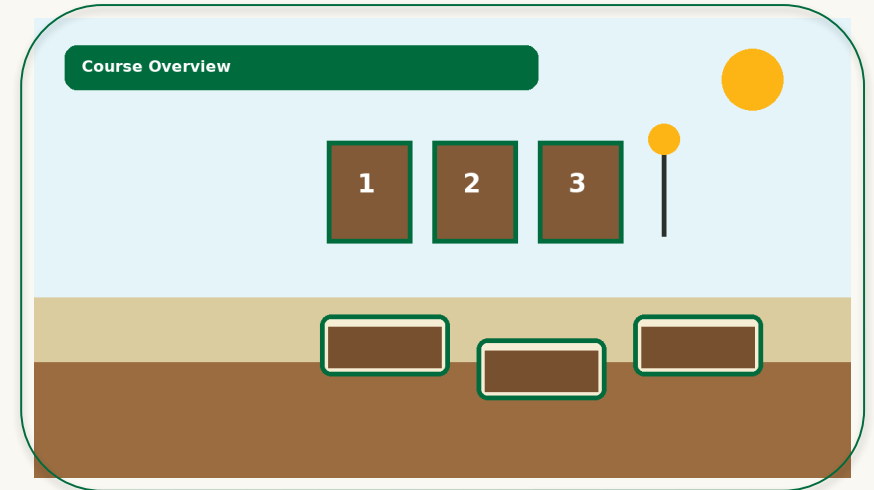
- LIFEWS AgriShine™ GrowMeal™ JSS 1-3 Teacher Training
- Audience: JSS 1-3 teachers, agricultural-science educators and garden facilitators
- Format: Self-paced online course + supervised practical/laboratory orientation
- Recommended duration: 3.5 hours online + 2 hours practical
- JSS learning progression: formulate -> test -> measure -> analyze -> manage



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the garden/laboratory system; teach all ten technical topics accurately; guide students in measurement, comparison and data collection; connect agricultural science with mathematics and systems reasoning; supervise practical work safely; maintain credible records; interpret evidence without overstating conclusions.
- Core materials: teacher guide, system/plot map, approved tools and irrigation equipment, labels, clipboards, rulers/tapes, graduated containers, meters/sensors where relevant, data sheets, clean sample/harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical session, and submit a lesson, measurement protocol, experiment or systems investigation appropriate to the course.



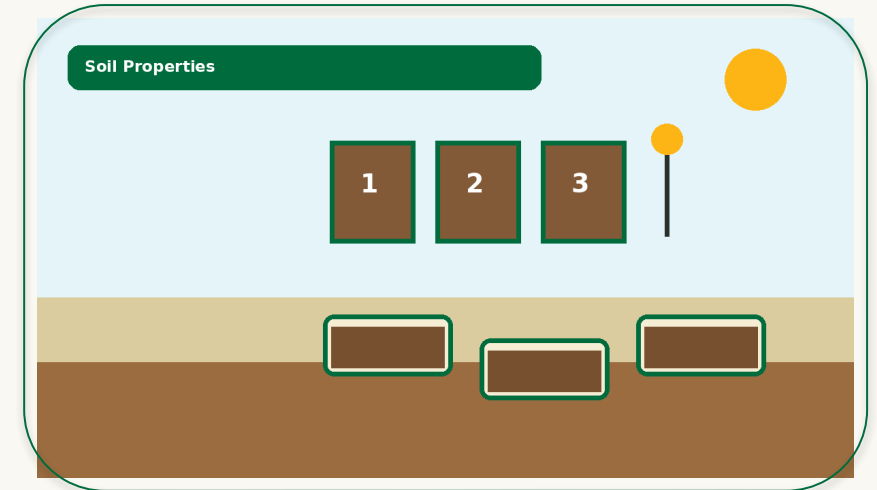
TEACHER PRACTICE

- Training sequence: QUESTION -> PLAN -> IMPLEMENT -> MEASURE -> ANALYZE -> DECIDE.

Soil Properties

Teacher knowledge, technical practice and student supervision

- Introduce physical, chemical and biological properties as interacting dimensions of soil condition.
- Observe color, aggregation, root presence, moisture and other safe field indicators without treating them as complete diagnoses.
- Connect properties to root growth, water movement and nutrient availability.
- Use laboratory or field tests where available for quantitative information.
- Record site and depth for every soil observation.



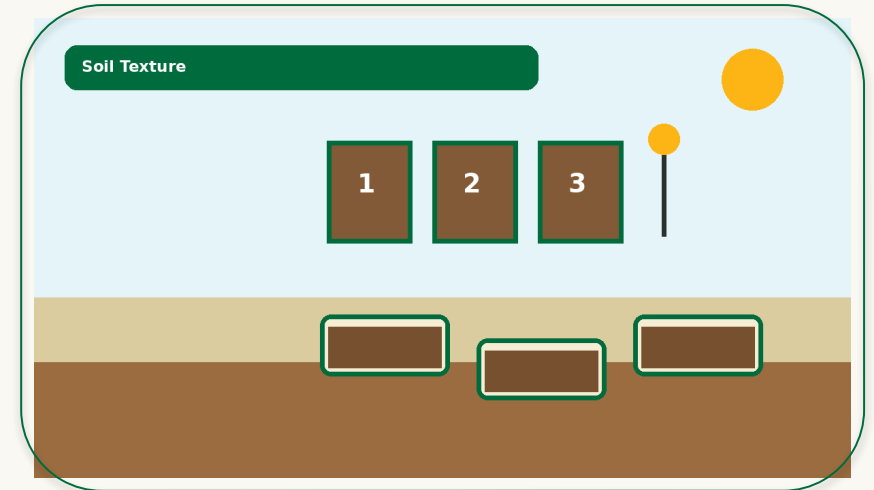
TEACHER PRACTICE

- Prepare a soil-property observation checklist.

Soil Texture

Teacher knowledge, technical practice and student supervision

- Define texture as the relative proportions of sand, silt and clay.
- Use a safe hand-texturing demonstration or settling activity where appropriate.
- Explain that texture is relatively stable compared with moisture or organic matter.
- Relate texture to infiltration and water-holding tendencies without oversimplifying.
- Do not classify contaminated or unknown soils by hand.



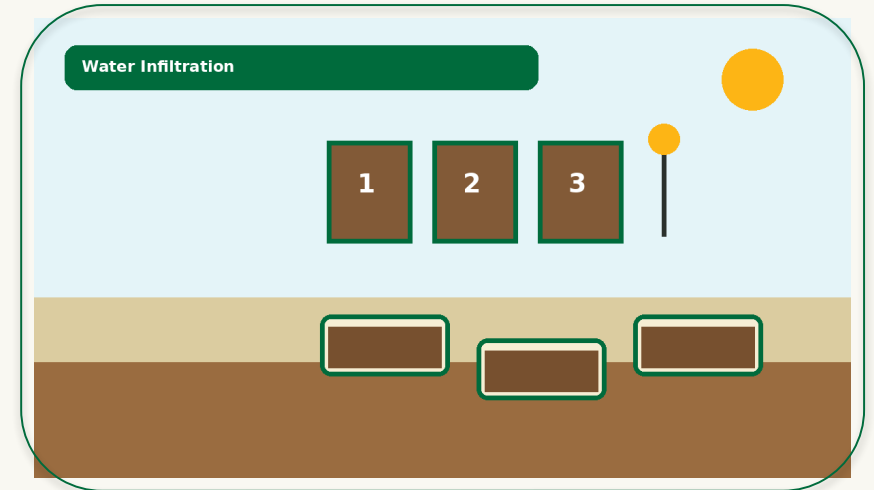
TEACHER PRACTICE

- Compare two safe soil samples using a consistent texture procedure.

Water Infiltration

Teacher knowledge, technical practice and student supervision

- Define infiltration as entry of water into the soil surface.
- Use a simple ring or bounded-area test with a measured water volume.
- Record time and test location.
- Repeat at more than one location because soil conditions vary.
- Discuss how compaction, surface cover and soil structure may influence results.



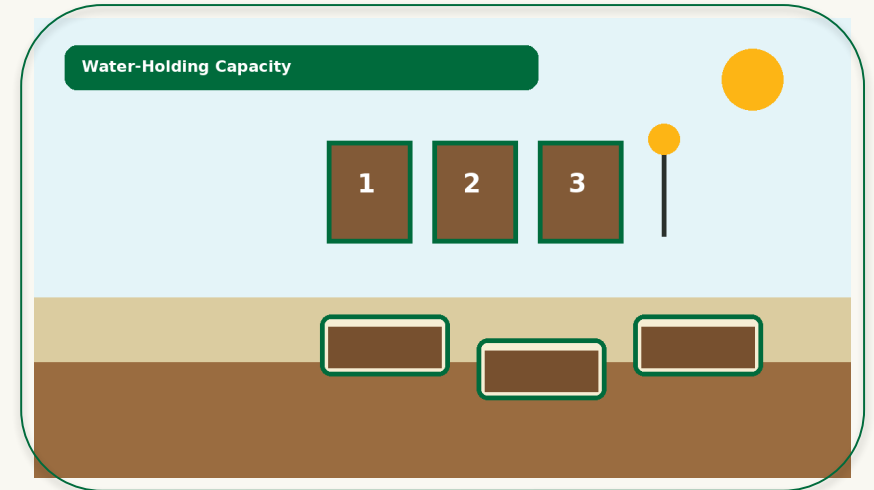
TEACHER PRACTICE

- Conduct and record a simple infiltration test.

Water-Holding Capacity

Teacher knowledge, technical practice and student supervision

- Explain water-holding capacity as the amount of water a soil or medium can retain under defined conditions.
- Compare equal masses/volumes of safe media using a controlled classroom demonstration.
- Measure water added and drainage collected where feasible.
- Keep sample amount and procedure consistent.
- Distinguish a demonstration from a formal laboratory determination.



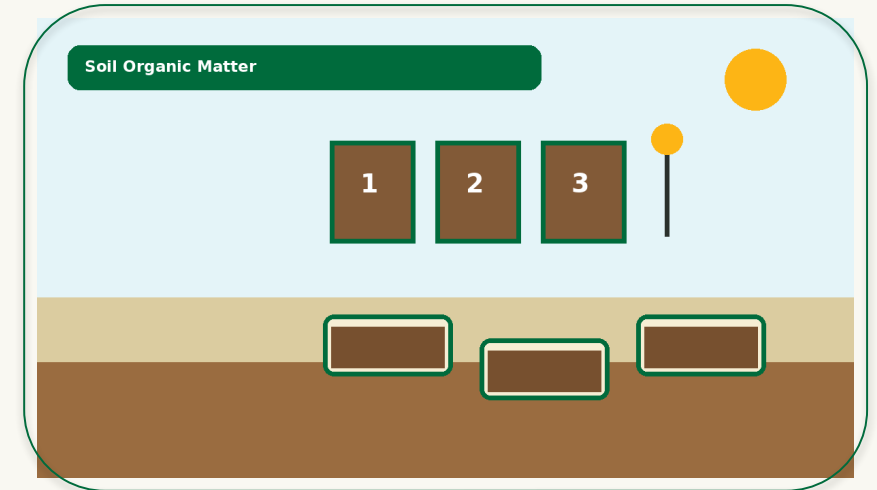
TEACHER PRACTICE

- Prepare a controlled water-retention comparison.

Soil Organic Matter

Teacher knowledge, technical practice and student supervision

- Explain soil organic matter as carbon-containing material in various stages of decomposition within soil.
- Discuss roles in aggregation, nutrient cycling and water relations without promising fixed effects.
- Use mature compost as an example of an organic amendment, not as identical to native soil organic matter.
- Record amendment history when comparing soils.
- Avoid judging organic matter content from darkness alone.



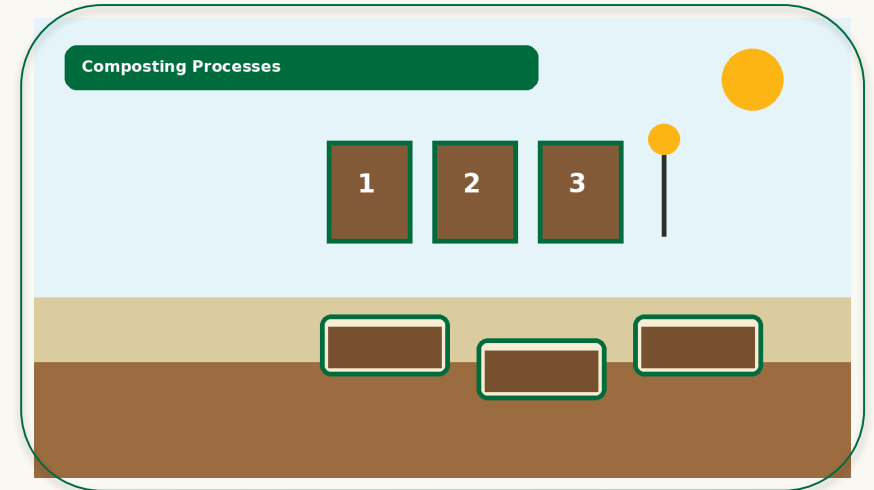
TEACHER PRACTICE

- Prepare an organic-matter concept map linked to soil functions.

Composting Processes

Teacher knowledge, technical practice and student supervision

- Review decomposition, aeration, moisture, temperature and material balance.
- Explain microbial activity as the driver of compost transformation.
- Monitor active compost using approved tools and safe access.
- Record major additions and turning events.
- Keep pupils away from fresh manure, hazardous waste and unsafe hot-pile handling.



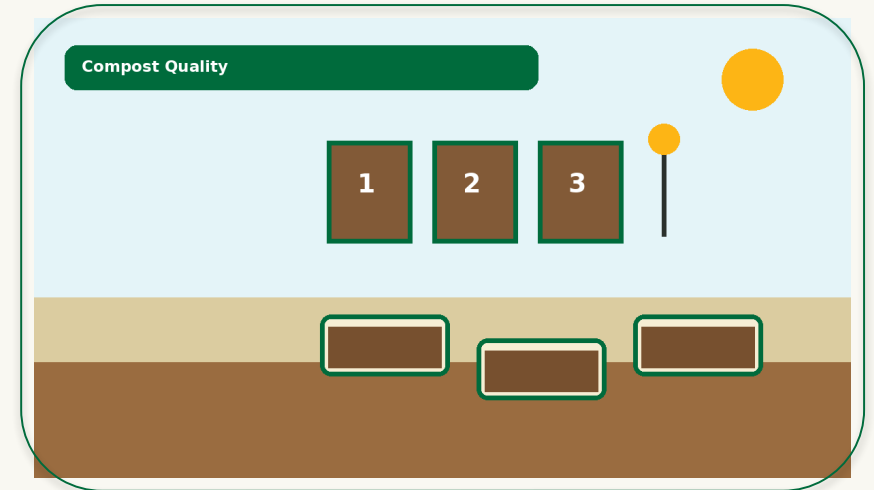
TEACHER PRACTICE

- Create a compost process monitoring schedule.

Compost Quality

Teacher knowledge, technical practice and student supervision

- Assess maturity, odor, texture, contamination and consistency using school-approved criteria.
- Explain that mature-looking compost can still vary in nutrient content and salinity.
- Use laboratory testing where needed for higher-confidence quality information.
- Reject compost containing plastics, glass or hazardous contaminants.
- Store finished compost to prevent recontamination.



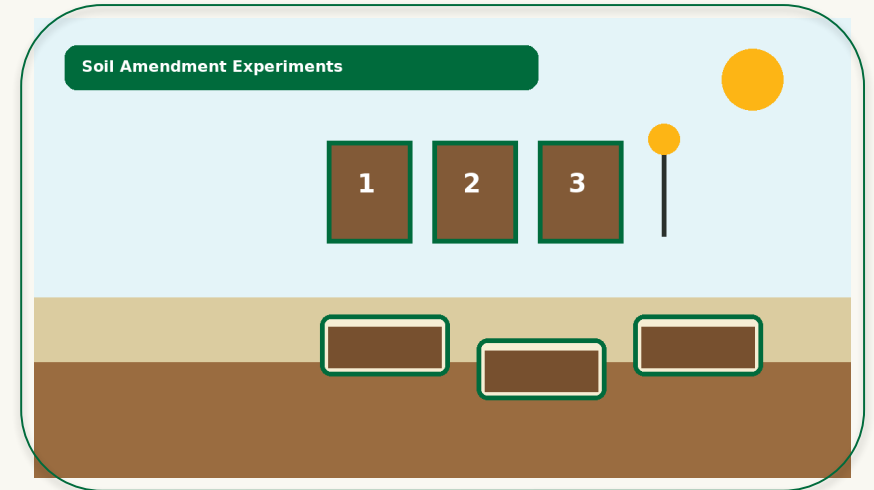
TEACHER PRACTICE

- Develop a compost acceptance checklist.

Soil Amendment Experiments

Teacher knowledge, technical practice and student supervision

- Compare approved amendment treatments such as different mature-compost rates against a reference soil.
- Define treatment rates clearly and use replication.
- Keep crop, container/plot size, watering and measurement methods similar.
- Measure plant and soil responses rather than relying on appearance alone.
- Avoid unsafe or excessive amendment rates.



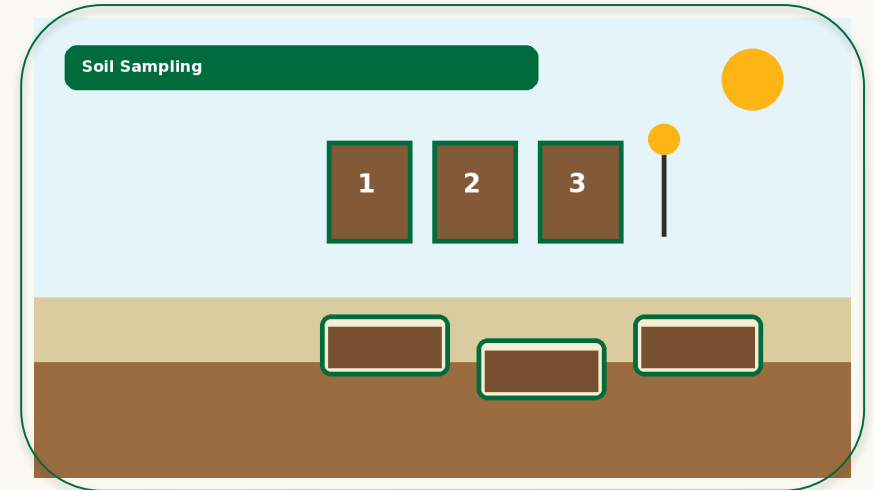
TEACHER PRACTICE

- Design a replicated compost-amendment experiment.

Soil Sampling

Teacher knowledge, technical practice and student supervision

- Define sampling objective, depth, number of subsamples and labeling before collection.
- Use clean tools and a systematic or representative pattern.
- Create composite samples only when appropriate to the objective.
- Avoid atypical spots unless intentionally sampled separately.
- Document block, depth, date and recent amendments.



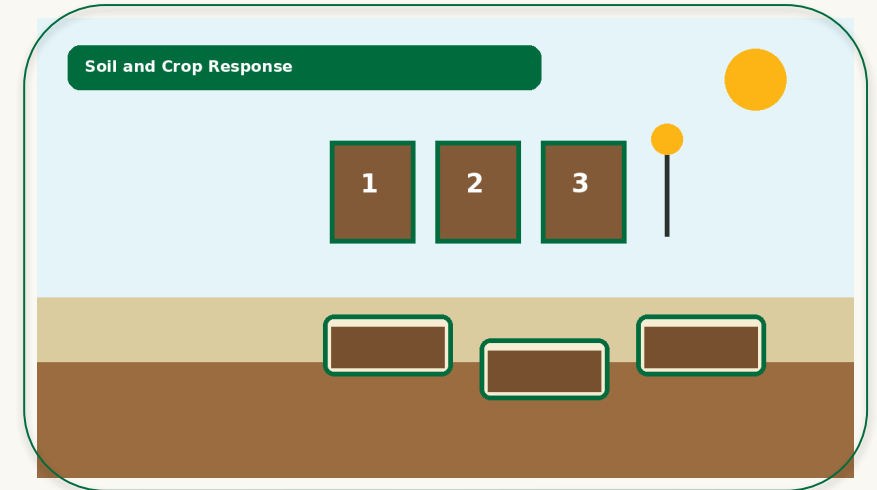
TEACHER PRACTICE

- Prepare a field soil-sampling SOP.

Soil and Crop Response

Teacher knowledge, technical practice and student supervision

- Compare soil indicators with crop emergence, growth, health observations and yield.
- Use time-series or treatment data where possible.
- Recognize that crop response reflects multiple interacting factors.
- Avoid inferring a specific nutrient deficiency or soil cause from one symptom.
- Use findings to refine soil-management decisions.



TEACHER PRACTICE

- Completion task: prepare a soil-to-crop evidence report.